Dealer-Gamma Feedback and Local Volatility Cycles: A Fixed-Equilibrium Bifurcation Model and a Pre-Extraction Identification Protocol

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Abstract

We study a physical-measure local model in which detrended log price, square-root variance, and an exponentially filtered price signal form a closed feedback loop. The model fixes the reference equilibrium while the coupling varies and makes the variance drift point inward at zero. For a general three-times differentiable book functional, we state all Routh–Hurwitz side conditions for a local Hopf bifurcation, derive the exact eigenvalue crossing speed, and evaluate the first Lyapunov coefficient. For a centered Gaussian density of latent signed dealer positions in log moneyness, the Hopf determinant is quadratic in the coupling. The first positive local Hopf point of a deliberately constructed, non-calibrated configuration has $\kappa^\star = 31.4933\,\mathrm{yr}^{-1}$, $\omega^\star = 47.1186\,\mathrm{rad\,yr}^{-1}$, and $\ell_1 = -6.2888$. Integration of the actual nonlinear book gives an attracting cycle, and its local amplitude scales approximately as $\{(\kappa - \kappa^\star)/\kappa^\star\}^{0.509}$ over the closest numerical grid. A gross-normalized mixture audit shows a more qualified conclusion: nearby mixtures preserve the supercritical Hopf, but dispersed or offsetting books can make it subcritical, and reversing the canonical signed orientation removes every positive local Hopf root. Thus the paper proves a mechanism possibility, not robustness of one market regime or evidence that SPX is near the threshold. Public open interest identifies neither participant nor side, so open-interest-weighted signed gamma is a convention, not observed dealer inventory. We therefore pre-specify a registered-horizon, sign-agnostic predictive protocol and its measurement, timing, attrition, multiplicity, and interpretation rules before proprietary data access. No registered market dataset is analysed in this paper.

1 Introduction

Delta hedging turns an option position into state-dependent demand for its underlying. When the intermediary sector is net long gamma, rebalancing is locally contrarian; when it is net short gamma, it is locally trend-following. This observation motivates a possible feedback channel from option inventories to returns and volatility (Frey and Stremme, 1997; Sircar and Papanicolaou, 1998; Platen and Schweizer, 1998). Demand-based option pricing establishes that intermediary balance sheets and end-user demand can affect *option prices* (Gârleanu et al., 2009); it does not, by itself, identify a drift term for the underlying. Our spot-feedback equation should therefore be read as a reduced-form hypothesis to be tested, not as a consequence of that paper.

The purpose of this paper is deliberately narrower than a structural model of the entire option market. We ask whether a smooth, three-state local model can admit an endogenous oscillatory instability without moving the equilibrium as the coupling parameter changes, without allowing the variance feedback to push through zero, and without confusing a public open-interest proxy with signed dealer inventory. These requirements matter because violating any one of them can manufacture an apparent bifurcation or an apparent empirical test. The paper does not estimate dealer positions, derive the drift from an intermediary optimization problem, or claim that observed volatility clustering is a deterministic limit cycle.

The paper makes four contributions.

1. We state a physical-measure model for detrended log-price deviations. Centering the dealer-book functional pins $(X, v, \chi) = (0, \theta_v, 0)$ for every coupling κ. Multiplying the memory feedback in the variance drift by v leaves the boundary drift equal to $\kappa_v \theta_v > 0$.

2. We derive the complete cubic Routh–Hurwitz test, including all positivity side conditions and an exact expression for transversality. The result is local: absence of a Hopf root is not a global-stability theorem, and a subcritical Hopf alone does not prove hysteresis.

3. We specialize the dealer book to a centered Gaussian signed-position density. The Hopf equation becomes a quadratic in κ, while the first Lyapunov coefficient is evaluated from analytic second and third derivatives. The numerical example integrates this actual nonlinearity rather than a separately chosen cubic normal form. We then audit normal-form scaling, numerical tolerances, initial conditions, one-at-a-time parameter changes, the book-shape plane, and signed Gaussian mixtures.

4. We state what can and cannot be learned from the future WRDS exercise. Because public open interest does not identify dealer sign, the revised protocol is a registered-horizon proxy study, not a causal or model-confirming dealer-inventory test.

Several claims in earlier working drafts are deliberately absent. An earlier draft replaced the state-dependent diffusion by the affine additive surrogate $\mathrm{d}Z = JZ\,\mathrm{d}t + \varepsilon\Sigma\,\mathrm{d}W$ and then estimated a non-zero stochastic shift from forced trajectories. In that surrogate the same-noise derivative cocycle is exactly e^{Jt}, so its correction is zero. This does not solve the stochastic variational equation of (1)–(2), whose diffusion derivatives are non-zero; no stochastic threshold claim is made here. A Hawkes branching ratio near one is not a theorem that this three-dimensional system is near an oscillatory bifurcation. Neither a mean-field analogy nor an information statistic identifies the dealer-gamma mechanism. These observations mark the boundary between what is proved and what remains conjectural.

The rest of the paper is organized as follows. Section 2 positions the result relative to hedging-feedback, option-demand, endogenous-cycle, and empirical gamma work. Section 3 states the economic object and measurement distinction. Section 4 gives the local analysis, and sections 5 and 6 report the nonlinear and robustness audits. Section 7 fixes the future empirical protocol. The appendices derive the Gaussian book, list all derivatives used in ℓ_1, and record the numerical and empirical audit trails.

2 Position in the literature

Hedging feedback. The closest theoretical ancestors derive price or volatility feedback from the demand of dynamic hedgers. Frey and Stremme (1997), Sircar and Papanicolaou (1998), and Platen and Schweizer (1998) show that replication demand need not be price-taking when the hedging sector is large. Those models supply the economic reason that gamma-sensitive rebalancing can alter the underlying process. More recent work studies explicit impact, optimal hedging, and market quality. Anderegg et al. (2022) combine a permanent-impact model with reconstructed FX option-market-maker exposures; Buis et al. (2024) add a dynamic hedger to a simulated limit-order market. This paper does not improve their microfoundation. Its contribution on this axis is a fixed-equilibrium local bifurcation calculation with a positive-variance state and a fully differentiated nonlinear book.

Option demand and dealer balance sheets. Demand-based option pricing links end-user demand and intermediary constraints to option prices (Gârleanu et al., 2009). It does not imply the particular physical-measure spot drift used below. We therefore treat κg as a reduced-form price-pressure channel whose scale includes market depth, hedge frequency, and the fraction of positions actively rebalanced. Any statement that our drift is "derived from" demand-based option pricing would be too strong.

Endogenous financial cycles. Hopf and Neimark–Sacker bifurcations are established mechanisms in heterogeneous-agent asset-pricing models (Brock and Hommes, 1998; He et al., 2009; Brock et al., 2009). Moving-average rules provide a memory channel in continuous-time financial models (He and Zheng, 2010). These precedents show that endogenous oscillation is not itself novel. The distinct object here is the combination of a filtered price state, a variance state, and a centered signed option-book functional for which the coupling can be varied without changing the reference equilibrium.

Empirical gamma positioning. Empirical studies generally require an inventory reconstruction. Barbon and Buraschi (2021) and Dim et al. (2025) study gamma-related price or volatility effects using positioning assumptions or market-maker inventory measures, while Baltussen et al. (2021) connect intraday momentum to gamma-sensitive hedging demand across futures markets. Vasquez et al. (2025) uses participant-tagged information to estimate market-maker positions. The contrast is central to our design: a convention applied to public open interest is not observationally equivalent to participant-tagged inventory. Evidence from FX trade-repository data (Anderegg et al., 2022) is informative about the hedging channel but does not solve the measurement problem for SPX public open interest.

Scope of the contribution. Relative to these four strands, we do not claim gamma feedback per se, financial Hopf bifurcations per se, or signed dealer inventory from public data. The contribution is the auditable conjunction: a centered and boundary-compatible local system; complete cubic side conditions and transversality; analytic derivatives through third order; direct nonlinear and mixture checks; and an empirical protocol whose attainable estimand is kept separate from the latent theoretical object.

3 Economic object, measure, and state variables

All dynamics are under the physical measure $\mathbb{P}$. Let F_t be a positive, predictable reference level that absorbs the baseline expected return, and define the local detrended deviation

$$X_t = \log(S_t/F_t).$$

The analysis concerns a neighbourhood of $X = 0$ over a horizon on which the option book is frozen. It is not a global model for the price level. The proposed dynamics are

$$\mathrm{d}X_t = \left[-\delta X_t - \tfrac{1}{2}(v_t - \theta_v) + \kappa g(X_t, \chi_t, v_t)\right] \mathrm{d}t + \sqrt{v_t}\, \mathrm{d}W_t^S, \tag{1}$$

$$\mathrm{d}v_t = \left[\kappa_v(\theta_v - v_t) + \gamma v_t \chi_t\right] \mathrm{d}t + \xi \sqrt{v_t}\, \mathrm{d}W_t^v, \tag{2}$$

$$\mathrm{d}\chi_t = \alpha(\beta X_t - \chi_t)\, \mathrm{d}t, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\, \mathrm{d}t. \tag{3}$$

We take $\delta, \kappa_v, \theta_v, \alpha, \beta > 0$, $\gamma, \kappa, \xi \geq 0$, and $\rho \in [-1, 1]$. Here $\delta, \kappa_v, \alpha, \gamma$, and κ have units yr^{-1}; X, χ, β, and g are dimensionless; and v and θ_v are annualized instantaneous variances under the convention that a year is the time unit. The volatility-of-variance coefficient ξ has units $\mathrm{yr}^{-1/2}$ and ρ is dimensionless. The term $-\frac{1}{2}(v - \theta_v)$ is the Itô correction relative to the reference variance. In the nested limit $\delta = \gamma = \kappa = 0$, after restoring the predictable reference drift, (X, v) has Heston-type log-return and CIR-variance dynamics (Heston, 1993). Setting $\kappa = 0$ alone does *not* remove the $X \to \chi \to v$ channel, so we do not call that restriction Heston.

The variance feedback is proportional to v. At the boundary,

$$b_v(0, \chi) = \kappa_v \theta_v > 0, \qquad \sigma_v(0) = 0,$$

independently of χ. Thus the added drift is compatible with the non-negative variance state space. The usual Feller inequality $2\kappa_v\theta_v \geq \xi^2$ is a separate sufficient boundary condition for the unperturbed CIR backbone (Feller, 1951); we do not use it in the deterministic bifurcation theorem. Numerical simulation of the stochastic system should still use a positivity-preserving scheme rather than silently evaluating $\sqrt{v}$ at a negative Euler iterate (Lord et al., 2010).

Assumption 1 (Local frozen-book experiment)**.** *Over the theoretical horizon, F_t is predictable, the position density and effective maturity are frozen, and g is C^3 on an open neighbourhood of $(0, 0, \theta_v)$. Parameters other than κ are held fixed when the bifurcation parameter varies. The deterministic orbit considered below remains inside that neighbourhood and satisfies $v > 0$.*

This assumption defines a local comparative-static experiment, not a claim that an actual SPX book stays constant for weeks. Freezing positions isolates the feedback geometry; endogenous inventory formation would add states and can change both the threshold and the bifurcation type.

3.1 From hedge demand to a reduced-form state feedback

For intuition, suppose an intermediary holds a signed option density $q(k)$ and offsets its option delta with underlying position

$$h(S, v) = -\int q(k)\Delta^{BS}(S, K(k), v, T)\, \mathrm{d}k.$$

Holding implied variance and the book fixed over an infinitesimal price change gives

$$\frac{\partial h}{\partial S} = -\int q(k)\Gamma^{BS}(S, K(k), v, T)\, \mathrm{d}k.$$

Thus a long-gamma intermediary sells underlying after a price increase, whereas a short-gamma intermediary buys. This sign calculation explains why a signed gamma-weighted book is economically relevant. It does *not* derive κg in (1): converting hedge demand into a calendar-time drift requires assumptions about rebalancing speed, temporary and permanent impact, other liquidity suppliers, and simultaneous changes in implied volatility. Those objects are compressed into κ and the chosen state dependence. The reduced form is consequently falsifiable but not structurally identified. In particular, g is a state-dependent pressure proxy, not the hedge trade $\mathrm{d}h$ itself: the model does not solve the simultaneous price-impact equation generated by $\partial h/\partial S$, other order flow, and market clearing.

The diffusion coefficients do not enter the deterministic Hopf threshold because the threshold is a property of the drift skeleton. They remain economically important: the full stochastic system has state-dependent diffusion, its paths do not converge to a deterministic cycle, and the deterministic threshold need not equal a stochastic bifurcation threshold.

3.2 Latent signed positions are not open interest

Let $q(k)$ denote a latent density of *signed dealer positions* at log moneyness $k = \log(K/F_\star)$. A positive value means the dealer sector is long the option under the chosen normalization. Every outstanding option has a long and a short side. Public open interest records the number of outstanding contracts; it does not label the dealer side or even the identity of either holder (Options Industry Council, 2025). Consequently, an empirical open-interest grid cannot be substituted for q without an additional, untestable sign assignment. Recent work that estimates option-market-maker gamma uses proprietary participant-tagged trade data precisely to recover positions (Vasquez et al., 2025).

Proposition 1 (Non-identification of signed dealer inventory from aggregate OI)**.** *Fix contract-level public open interest n_i. Without participant labels or restrictions on which participant type holds each side, the signed dealer position vector is not identified: there exist allocations with the same $(n_i)_i$ and opposite dealer signs.*

Proof. Each of the n_i outstanding contracts contributes one long and one short position. An allocation assigning dealers to the long side and non-dealers to the short side has the same public count as the allocation obtained by swapping those participant labels. Dealer net position changes sign while open interest is unchanged. Applying the construction contract by contract proves that the observation map is not injective. □

This is a data-content result, not a small-sample problem: more dates of the same public OI field do not identify dealer sign without additional information or restrictions.

For theory we first define a positive unit-mass Gaussian shape over fixed log moneyness,

$$\varphi_q(k) = \frac{1}{\sqrt{2\pi\sigma_q^2}} \exp\left\{-\frac{(k-\mu_q)^2}{2\sigma_q^2}\right\},$$

and integrate Black–Scholes spot gamma (Black and Scholes, 1973). Up to gross position mass and an impact scale absorbed by κ, the integral is

$$\mathcal{G}(X,v) = \frac{\exp(-q_d T - X)}{\sqrt{2\pi\tau^2(v)}} \exp\left[-\frac{(X-m(v))^2}{2\tau^2(v)}\right], \tag{4}$$
$$\tau^2(v) = \sigma_q^2 + vT, \qquad m(v) = \mu_q - (r - q_d + v/2)T.$$

The functional used in (1) is centered and normalized:

$$g(X,\chi,v) = s\,\frac{\mathcal{G}(X,v) - \mathcal{G}(0,\theta_v)}{\mathcal{G}(0,\theta_v)}, \qquad s \in \{-1,+1\}. \tag{5}$$

The sign s is a latent dealer-position orientation, not an open-interest statistic. More general C^3 functionals, including mixtures and a dependence on χ, fit the theorem below provided $g(0,0,\theta_v) = 0$. The Gaussian specialization is intentionally static in χ so that its assumptions are auditable. Section A derives (4) by completing the square rather than treating it as an assumed response surface.

Equation (4) also makes a strong low-dimensional closure: the same state v is inserted as one flat Black–Scholes variance for every strike at the effective maturity. The physical-measure variance state does not generally equal option-implied variance, and a volatility smile would add further states or a surface. This closure is used only to obtain an analytic witness; it is not a pricing or risk-premium claim.

Remark 1 (What centering does)**.** *Centering removes the level of the book functional at the reference state, not its local price and variance sensitivities. It is equivalent to absorbing a constant pressure into the predictable reference drift. It avoids changing that drift as κ varies; hence the bifurcation parameter traces one family of vector fields with a common equilibrium.*

Proposition 2 (Positive normalization invariance)**.** *Let $\widetilde{g} = cg$ for a constant $c > 0$ and $\widetilde{\kappa} = \kappa/c$. Then the vector field, Hopf frequency, crossing direction, and first-Lyapunov sign at corresponding parameter values are unchanged. In particular, the numerical magnitude of $\kappa^\star$ cannot be interpreted without the normalization of g.*

Proof. Every occurrence of the book in the vector field is through the product κg. Consequently all state derivatives of the drift at $(\widetilde{\kappa}, \widetilde{g})$ equal those at (κ, g). The Jacobian and Taylor tensors B, C are identical, so all stated local quantities agree. □

4 Fixed-equilibrium local Hopf bifurcation

Remove the Brownian terms and write the deterministic drift as $f(X, v, \chi; \kappa)$. Because $g(0,0,\theta_v) = 0$,

$$z_\star = (0, \theta_v, 0) \tag{6}$$

is an equilibrium for every $\kappa \geq 0$. Put

$$G_X = \partial_X g(z_\star), \qquad G_v = \partial_v g(z_\star), \qquad G_\chi = \partial_\chi g(z_\star).$$

The Jacobian is

$$J(\kappa) = \begin{pmatrix} a & b & d \\ 0 & -\kappa_v & c \\ m & 0 & -\alpha \end{pmatrix}, \qquad \begin{matrix} a = -\delta + \kappa G_X, & b = -\frac{1}{2} + \kappa G_v, & d = \kappa G_\chi, \\ c = \gamma\theta_v, & m = \alpha\beta. & \end{matrix} \tag{7}$$

For $\det(\lambda I - J) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$, direct expansion gives

$$c_2 = \kappa_v + \alpha - a, \tag{8}$$

$$c_1 = \kappa_v\alpha - a(\kappa_v + \alpha) - dm, \tag{9}$$

$$c_0 = -a\kappa_v\alpha - bcm - dm\kappa_v. \tag{10}$$

The equilibrium is linearly asymptotically stable exactly when $c_2 > 0$, $c_1 > 0$, $c_0 > 0$, and

$$H(\kappa) := c_1(\kappa)c_2(\kappa) - c_0(\kappa) > 0. \tag{11}$$

Theorem 1 (Local Hopf criterion)**.** *Suppose f is C^3 jointly in state and κ near $(z_\star, \kappa^\star)$. If*

$$c_2(\kappa^\star) > 0, \quad c_1(\kappa^\star) > 0, \quad c_0(\kappa^\star) > 0, \quad H(\kappa^\star) = 0, \quad H'(\kappa^\star) \neq 0, \tag{12}$$

then $J(\kappa^\star)$ has a simple pair $\lambda_\pm = \pm i\omega^\star$ and a negative real eigenvalue, where

$$\omega^\star = \sqrt{c_1(\kappa^\star)}, \qquad \lambda_3 = -c_2(\kappa^\star).$$

The crossing speed is

$$\left.\frac{\mathrm{d}\,\mathrm{Re}\,\lambda_+}{\mathrm{d}\kappa}\right|_{\kappa^\star} = -\frac{H'(\kappa^\star)}{2\{c_1(\kappa^\star) + c_2(\kappa^\star)^2\}}. \tag{13}$$

If, in addition, the first Lyapunov coefficient $\ell_1 \neq 0$, a non-degenerate local Hopf bifurcation occurs. In the convention of Kuznetsov (2004), $\ell_1 < 0$ gives stable small-amplitude periodic orbits on the side where the equilibrium pair has positive real part; $\ell_1 > 0$ gives unstable orbits on the opposite side.

Proof. At $H = 0$ with the stated positivity conditions, the cubic factors as $(\lambda + c_2)(\lambda^2 + c_1)$, proving the eigenvalue statement. Implicitly differentiate $P(\lambda, \kappa) = 0$ at $\lambda = i\sqrt{c_1}$ and take real parts. Substitution of $c_0 = c_1 c_2$ yields (13). The non-zero derivative and $\ell_1 \neq 0$ are the transversality and non-degeneracy conditions of the classical Hopf theorem (Liu, 1994; Kuznetsov, 2004). □

This theorem says nothing about equilibria or invariant sets outside a neighbourhood of $z_\star$. In particular, a negative discriminant for a later quadratic only rules out that local Hopf mechanism. It does not prove stability for all couplings, exclude a real eigenvalue crossing, or exclude a distant cycle. Likewise, a subcritical local Hopf gives an unstable small orbit; hysteresis additionally requires global coexistence and fold-of- cycles structure.

4.1 Static book: a quadratic threshold equation

For (5), $G_\chi = 0$. Define

$$A = \alpha + \kappa_v, \qquad M = \alpha\kappa_v, \qquad L = (\gamma\theta_v)(\alpha\beta).$$

The product L is the linear gain around the directed loop $X \to \chi \to v \to X$; the last link is supplied by b in (7).

Proposition 3 (A closed three-state loop is necessary in the static-book model)**.** *If $L = 0$, the eigenvalues are $a, -\kappa_v, -\alpha$ for every κ and no non-zero-frequency Hopf bifurcation is possible.*

Proof. When $\gamma\theta_v = 0$ or $\alpha\beta = 0$, one of the two off-diagonal links $v \leftarrow \chi$ and $\chi \leftarrow X$ vanishes. The Jacobian is triangular after a permutation, and its characteristic polynomial factors as $(\lambda - a)(\lambda + \kappa_v)(\lambda + \alpha)$. □

Substitution into (11) gives

$$H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0, \tag{14}$$

where

$$\begin{aligned} A_2 &= G_X^2 A, \\ A_1 &= G_v L - G_X A(A + 2\delta), \\ A_0 &= A(M + \delta A + \delta^2) - \tfrac{1}{2}L. \end{aligned} \tag{15}$$

At $\kappa = 0$, $c_2 = A + \delta > 0$, $c_1 = M + \delta A > 0$, and $c_0 = \delta M + L/2 > 0$. Hence the uncoupled equilibrium is stable exactly when

$$L < 2A(M + \delta A + \delta^2). \tag{16}$$

This inequality is checked in the numerical configuration; it is not automatic merely because every primitive parameter is positive.

Every positive real root must still be checked against $c_2 > 0, c_1 > 0, c_0 > 0$ and (13); blindly selecting the positive quadratic root is not a Hopf test. The derivatives G_X and G_v follow analytically from (4). Because $\mathcal{G}(0, \theta_v) > 0$, derivatives of the centered functional are simply s times the raw derivatives divided by that value.

To classify a valid root, let B and C be the symmetric bilinear and trilinear Taylor tensors of f at the equilibrium. For right and left eigenvectors q, p satisfying $Jq = i\omega q$, $J^\top p = -i\omega p$, and $\langle p, q \rangle = 1$, we evaluate

$$\begin{aligned}\ell_1 = \frac{1}{2\omega} \operatorname{Re} \Big\langle p, & C(q, q, \bar{q}) - 2B\{q, J^{-1}B(q, \bar{q})\} \\ & + B\{\bar{q}, (2i\omega I - J)^{-1}B(q, q)\} \Big\rangle.\end{aligned} \tag{17}$$

All derivatives of (4) through order three are analytic. Importantly, the variance equation also contributes $B_{v,v\chi} = B_{v,\chi v} = \gamma$; omitting this term changes ℓ_1. The repository tests the analytic book derivatives against symbolic differentiation, verifies the characteristic polynomial and eigenvalue factorization, and checks ℓ_1 against an independent finite-difference construction of B and C. The sign of ℓ_1 classifies the local Hopf. Its numerical magnitude depends on state and eigenvector normalization and is not an economic effect size. Sections B and C give the full derivative and computational audit trails.

5 Illustrative nonlinear validation

The configuration in table 1 is a constructive witness deliberately chosen to exhibit a well-separated first Hopf crossing in a readable daily-to-monthly regime while keeping the displayed orbit at positive variance. It is not fitted to SPX, drawn from a prior, or evidence that these parameter magnitudes are economically typical.

Object	Value	Interpretation
δ	$0.5\,\mathrm{yr}^{-1}$	local reference pull
κ_v, θ_v	$40\,\mathrm{yr}^{-1}$, 0.0625	variance reversion and level
α, β	$50\,\mathrm{yr}^{-1}$, 2	memory rate and loading
γ	$500\,\mathrm{yr}^{-1}$	variance–memory feedback
(μ_q, σ_q, T)	$(0.06, 0.20, 0.25\,\mathrm{yr})$	Gaussian signed book
r, q_d	$0, 0$	book-kernel carry inputs
s	$+1$	illustrative latent orientation

Table 1: Illustrative structural parameters. No row is an empirical estimate.

At the fixed equilibrium, the normalized book has $G_X = -0.0617978$ and $G_v = -2.0198878$. Enumerating both roots of (14) and applying every side condition gives the first valid point

$$\kappa^\star = 31.4932976\,\mathrm{yr}^{-1}, \quad \omega^\star = 47.1185670\,\mathrm{rad\,yr}^{-1}, \quad \ell_1 = -6.2888041. \tag{18}$$

The period $2\pi/\omega^\star = 0.1333484$ years is about 33.6 trading days under a 252-day convention, or 48.7 calendar days under ACT/365, and the crossing derivative is $0.2686294 > 0$. Therefore the local normal form predicts stable cycles just above the threshold. The loop gain is $L = 3125$, while the right side of the baseline-stability condition (16) is 368145. At the Hopf point, $(c_2, c_1, c_0) = (92.4462, 2220.1594, 205245.3288)$ and $H'(\kappa^\star) = -5784.3769$, so none of the classification conditions is numerically close to zero except the determinant condition being solved.

The quadratic also has a second valid positive local Hopf point at $\kappa^{\star\star} = 16860.8961\,\mathrm{yr}^{-1}$, with $\omega^{\star\star} = 309.5511\,\mathrm{rad\,yr}^{-1}$, $\mathrm{d}\,\mathrm{Re}\,\lambda_+/\mathrm{d}\kappa = -0.0020984$, and $\ell_1 = 17479.5557$. The pair therefore crosses back into the stable half-plane as κ increases through this second root, with an unstable small orbit on the locally stable side. We report it to make the quadratic audit complete; its coupling is more than 500 times the first threshold, and no nonlinear orbit or economic relevance is claimed in that remote regime.

Figure 1 checks more than the linear algebra. We integrate the actual nonlinear Gaussian-book drift at $1.02\kappa^\star$ from a small perturbation, discard 40 years of transient, and plot the final orbit. Over the retained trajectory, $X \in [-0.02814, 0.02358]$, $v \in [0.04453, 0.08187]$, and $\chi \in [-0.04297, 0.03310]$. Thus the displayed cycle remains in the physical variance domain and in a modest log-price neighbourhood. The long transient discard is a numerical convergence device, not an economic horizon.

The magnitude of κ depends on the normalization in (5). It cannot be compared with a per-dollar GEX coefficient without an explicit unit map. The illustration establishes internal possibility, not empirical proximity.

6 Robustness and numerical audit

A single orbit can verify implementation at one point but cannot establish that the result is structurally stable across book shapes or numerical choices. We therefore conduct five checks on the corrected centered model. None uses market data, and the parameter ranges are mechanical perturbations rather than calibrated confidence sets.

6.1 Local normal-form scaling and numerical convergence

For relative distances $\varepsilon_\kappa = (\kappa - \kappa^\star)/\kappa^\star$ in $\{0.000625, 0.00125, 0.0025, 0.005, 0.01, 0.02, 0.04, 0.06\}$, we integrate the actual Gaussian-book drift. The transient horizon is at least twelve linear growth time constants, $12/(\mathrm{Re}\,\lambda_+'(\kappa^\star)\kappa^\star\varepsilon_\kappa)$, with a floor of 80 years; only the last five simulated years are measured. The unusually long horizon is a convergence device necessitated by critical slowing near the threshold.

On the six points no more than two percent above threshold, a log–log regression of the half peak-to-peak X amplitude on ε_κ has slope 0.5089 and $R^2 = 0.99992$, close to the Hopf prediction 1/2. Including all points through six percent raises the slope to 0.5338, which documents departure from the leading local normal form rather than contradicting it. The measured period is 0.13341 years at 0.0625% above threshold, compared with the linear prediction 0.13335; it lengthens to 0.15667 by six percent. Thus both amplitude and period behave locally as the theorem predicts, while finite-distance corrections become visible.

At two percent above threshold, we tighten the DOP853 tolerance pair according to

$$(\mathrm{rtol}, \mathrm{atol}) : (10^{-8}, 10^{-10}) \longrightarrow (10^{-10}, 10^{-12}).$$

This changes the X amplitude by 1.94×10^{-11} relatively and does not change the reported period at the measurement resolution. Four initial conditions spanning both signs of X, variance offsets of

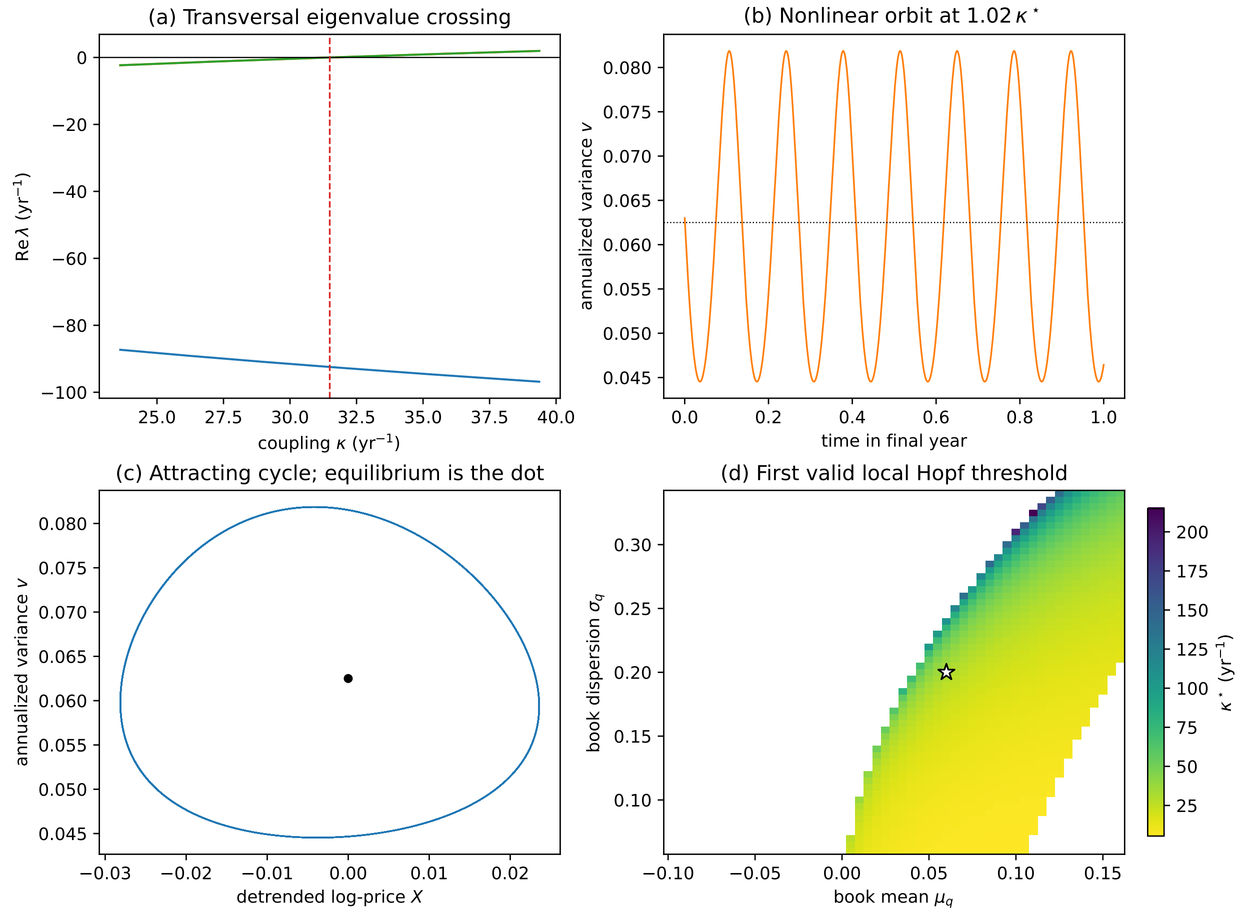

Figure 1: **Fixed-equilibrium Hopf validation using the actual nonlinear book.** (a) Real parts of all three eigenvalues; the complex pair crosses transversally at the dashed line. (b)–(c) Deterministic trajectory at $1.02\kappa^\star$ after the transient. The dotted line in (b) is θ_v and the dot in (c) is the fixed equilibrium. (d) First positive root satisfying all local Hopf side conditions when only (μ_q, σ_q) vary; the star is table 1. Blank cells mean "no valid local Hopf root," not "globally stable." Reproduction command: `python -m reflexive_options.experiments.centered_hopf_validation`.

± 0.003, and a non-zero memory perturbation converge to amplitudes with coefficient of variation 3.12×10^{-8} and periods with coefficient of variation 3.65×10^{-6}. These are numerical statements about the displayed attractor, not a proof of a global basin.

6.2 Book shape, mixtures, and what is not robust

Among the 2,025 cells in fig. 2(d), 747 have a valid positive local Hopf root; 722 are supercritical and 25 subcritical. This grid varies only two shape parameters with all dynamical parameters fixed, so its proportions are not probabilities. The sharp blank regions are economically important: a local Hopf is conditional on the shape and signed orientation of the latent book.

To move beyond a single Gaussian, let component j have positive mass w_j, orientation $s_j \in$

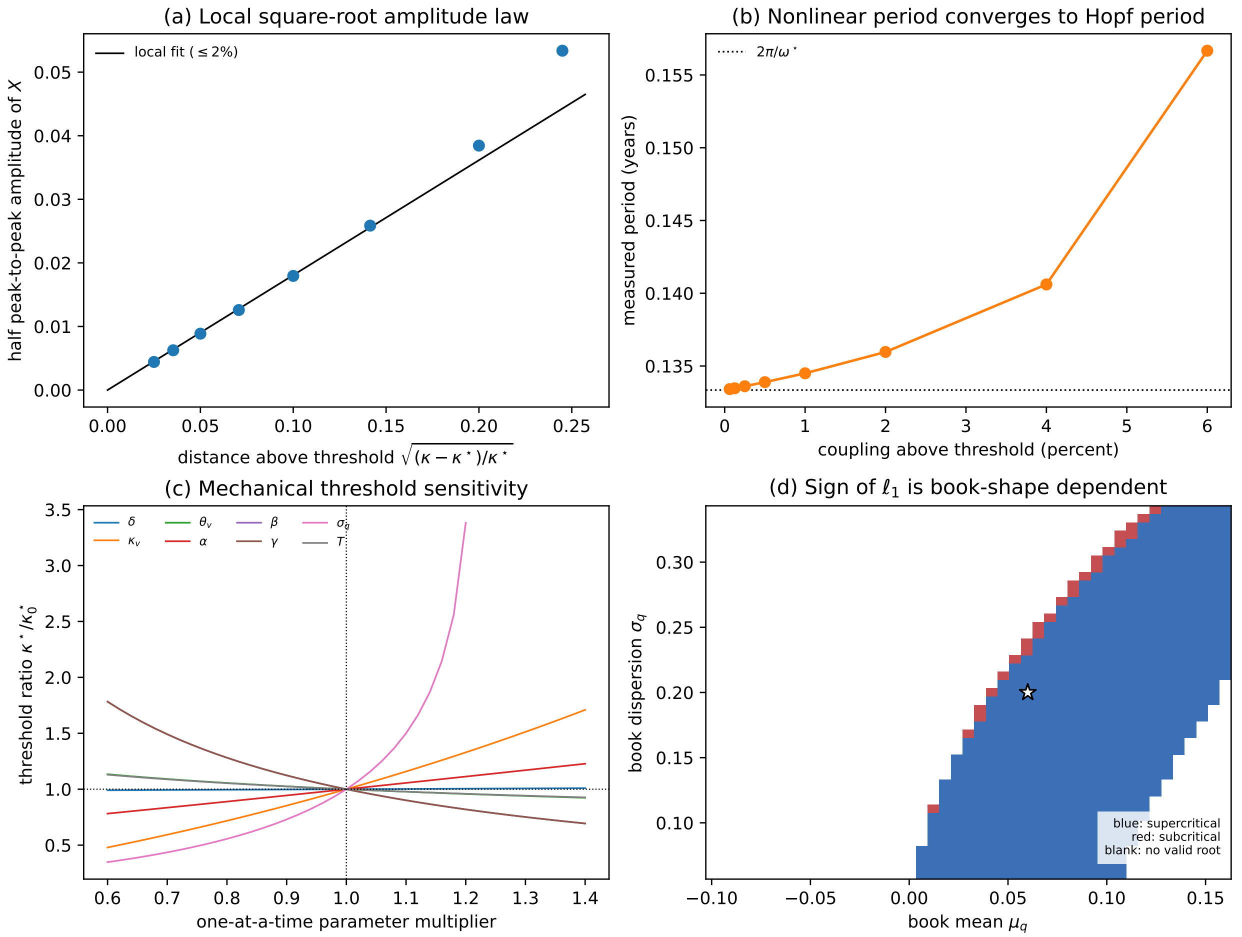

Figure 2: **Corrected-model robustness.** (a) The closest nonlinear cycles follow the square-root amplitude law. (b) Their periods approach $2\pi/\omega^\star$ at the threshold. (c) One-at-a-time perturbations from 60% to 140% of each canonical value show mechanical, not statistical, threshold sensitivity; gaps denote failure of a local Hopf side condition. (d) Over a 45×45 (μ_q, σ_q) grid, blue cells are supercritical, red cells subcritical, and blank cells have no positive root satisfying all local side conditions. Reproduction command: `python -m reflexive_options.experiments.centered_model_robustness`.

$\{-1, +1\}$, and raw kernel $\mathcal{G}_j$. We use

$$g_{\text{mix}}(X, v) = \frac{\sum_j w_j s_j \{\mathcal{G}_j(X, v) - \mathcal{G}_j(0, \theta_v)\}}{\sum_j w_j \mathcal{G}_j(0, \theta_v)}. \tag{19}$$

The denominator is gross equilibrium gamma. Normalizing by the signed net denominator would make the coupling explode mechanically when long and short components nearly cancel. Because differentiation is linear, all derivatives in eq. (17) are gross-normalized weighted sums of the component derivatives.

Table 2 prevents the strongest possible overstatement. The existence of a valid Hopf root survives several deformations, but supercriticality does not: second and third derivatives of the book can flip ℓ_1 without eliminating the imaginary-axis crossing. A subcritical result establishes only a local unstable orbit on the opposite side; it does not establish hysteresis or a large stable cycle. Reversing

Book	G_X	G_v	$\kappa^\star$	ℓ_1	Local classification
Canonical single	-0.0618	-2.0199	31.4933	-6.2888	supercritical
Nearby same-sign mixture	-0.1152	-2.0132	34.3891	-4.8143	supercritical
Dispersed same-sign mixture	-0.1514	-2.3480	30.2416	5.8498	subcritical
Offsetting-sign mixture	-0.1440	-1.5645	50.4759	27.8170	subcritical
Reversed canonical orientation	0.0618	2.0199	—	—	no valid positive root

Table 2: Gross-normalized mixture audit. Component definitions are fixed in the reproduction script. The examples are diagnostic constructions, not estimated books.

the canonical orientation produces no positive root satisfying all side conditions. Therefore neither the existence nor the type of the local bifurcation can be inferred from unsigned open interest.

6.3 Independent derivative check

Finally, we reconstruct every entry of B and C by central finite differences of the full drift at step 2×10^{-4}, without using the analytic book derivatives. The result is $\ell_1 = -6.288803983$, compared with -6.288804057 analytically: absolute difference 7.41×10^{-8} and relative difference 1.18×10^{-8}. This check is especially important because omitting the mixed derivative of $\gamma v \chi$ can preserve the Hopf threshold while changing its nonlinear classification.

7 What the future data can identify

No registered OptionMetrics, CRSP, or VIX dataset was loaded or analysed for this paper. The historical outcomes are nevertheless public and the named stress episodes were known when the protocol was written. This is therefore a retrospective pre-analysis plan made before proprietary-data extraction, not a prospectively blinded experiment. The timestamped original registration and its amendments remain in the repository. Amendments A13–A14 supersede the earlier event-selected directional GEX test and close the contract, source, and inference choices before extraction. Amendment A15 independently fixes market-calendar alignment, the outcome-session weekday control, and this registration language. The A13–A14 amendment file has SHA-256 `603e89366c0dbe49718e8c31f805d6f85d3c508e2e0ed4276a6310c80f5f9cd7` and OpenTimestamps receipt `pre_registration_amendments.md.ots`; calendar attestations are pending Bitcoin consolidation. A15 has SHA-256

`a5f694f99953d57563d4f17dc5646ef0b87452c45119ccbdc12fc90efd034a52`

and receipt `pre_registration_amendment_a15.md.ots`, also pending Bitcoin consolidation. The earlier records are preserved rather than overwritten.

The target of this exercise is an incremental predictive coefficient, not a structural parameter. Formally, it is the linear-projection coefficient of next-day log squared return on one standardized observable book summary conditional on the fixed controls below. It is not κ, G_X, G_v, a dealer inventory coefficient, or a causal response to an intervention on option demand. Making that estimand explicit prevents a statistically significant proxy coefficient from being mapped back into eq. (1) without an identification argument.

7.1 Primary proxy construction

On each eligible OptionMetrics date in the registered historical horizon from 3 January 2017 through 29 October 2024, retain only the verified SPX security, OSI roots SPX/SPXW, unadjusted 100-multiplier standard-settlement contracts, 1–365 calendar-day maturities, valid quotes and IV in $[0.01, 5]$, and $|\log(K/F)| \leq 0.50$. A parity forward is formed separately by date, expiration, and AM/PM settlement from the matched strike minimizing $|C_{mid} - P_{mid}|$, $F = K + e^{r\tau}(C_{mid} - P_{mid})$; carry is a flagged fallback when no pair survives. The spot is the OptionMetrics SPX close, r is linearly interpolated from its continuous zero curve, q is its index-dividend yield, and ACT/365 is used consistently. After these fixed filters, construct the unsigned gamma mass

$$U_t = \sum_i \mathrm{OI}_{i,t}\, \Gamma^{BS}_{i,t}\, S_t^2(0.01)(100), \tag{20}$$

together with three observable shape summaries. Write $u_{i,t} = \mathrm{OI}_{i,t}\Gamma^{BS}_{i,t} S_t^2(0.01)(100)$ and $k_{i,t} = \log(K_i/F_{i,t})$. The summaries are

$$B_t = \frac{\sum_{i\in C} u_{i,t} - \sum_{i\in P} u_{i,t}}{U_t}, \qquad M_t = \frac{\sum_i u_{i,t}k_{i,t}}{U_t}, \qquad D_t = \left\{\frac{\sum_i u_{i,t}(k_{i,t} - M_t)^2}{U_t}\right\}^{1/2}. \tag{21}$$

These are, respectively, call–put composition, gamma-weighted mean moneyness, and dispersion. They are unsigned properties of the public book and are not estimates of dealer holdings. Convention-based signed GEX series, including the original SqueezeMetrics construction (SqueezeMetrics, 2017), are reported only as secondary measurement sensitivities. Reversing a convention mechanically reverses its coefficient and is not robustness evidence.

The primary outcome is $Y_{t+1} = \log(r_{t+1}^2 + 10^{-8})$, where the immediately following CRSP trading-session log return is $r_{t+1} = \log(1 + \texttt{crsp.dsp500.sprtrn}_{t+1})$. VIX is the archived official Cboe daily close. For each standardized proxy, estimate a strictly forward regression

$$\begin{aligned} Y_{t+1} &= a + bQ_t + \phi_1 Y_t + \phi_5 \overline{Y}_{t-4:t} + \phi_{22}\overline{Y}_{t-21:t} \\ &\quad + \eta \log(\mathrm{VIX}_t) + \text{outcome-weekday and monthly-expiration controls} + e_{t+1}. \end{aligned} \tag{22}$$

Here Q_t is, in separate pre-specified regressions, $\log U_t$, B_t, M_t, or $\log(D_t + 10^{-6})$. Leads and lags are constructed on the complete ordered CRSP trading-session calendar before rows with a missing option proxy are removed. Thus a missing option date cannot compress the return horizon: $t + 1$ always denotes the immediately following CRSP session. Tuesday–Friday indicators refer to that outcome session, whose calendar identity is known at the close of t; the monthly-expiration indicator refers to the regressor session. The primary inferential target is incremental predictive association, $b = 0$ against a two-sided alternative. Report fixed-lag-5 Newey–West inference (Newey and West, 1987) and a moving-pairs block bootstrap (block 10, 2,000 draws, seed 42, 95% percentile interval) (Künsch, 1989). Benjamini–Hochberg at $q = 0.05$ is applied separately to the four HAC and four bootstrap p-values (Benjamini and Hochberg, 1995). A proxy is labelled robustly associated only when both adjusted p-values are below 0.05 and the bootstrap interval excludes zero; a one-family rejection is labelled method-sensitive, and every other case not detected. The same coefficient is used for both procedures, so the favourable method cannot be selected ex post.

The overlapping Y_t, five-day, and 22-day controls are intentionally fixed as a HAR-style predictive conditioning set. Their collinearity can inflate uncertainty without biasing OLS under full rank, so the primary table reports variance-inflation factors rather than removing whichever control is inconvenient. Proxy standardization uses one common option-date sample before outcome-based

Object	Locked source	Role and principal limitation
Option chain, OI, IV, spot	OptionMetrics IvyDB US	Constructs unsigned summaries; public OI contains no participant side
Forward, rates, dividends	Put–call parity by settlement group; OptionMetrics zero curve and index dividend	Carry is flagged fallback; all inputs are dated t
S&P 500 return	`crsp.dsp500.sprtrn`	Defines the next-CRSP-session outcome; not reconstructed from option spot
VIX	Archived official Cboe daily close	Predetermined control at close t; no interpolation or source substitution
Expiration calendar	Standard monthly SPX session	One fixed indicator; weekly-expiry definitions are not selected after results

Table 3: Pre-specified source map. Raw files, query text, retrieval times, row counts, date coverage, and SHA-256 hashes are frozen before outcomes are constructed.

listwise deletion. Because that transformation uses no future return and is affine, it changes coefficient units but not fitted values or the test of a zero coefficient.

The bootstrap resamples overlapping blocks of aligned (Y, X) rows and uses the same random block starts in all four regressions. The reported bootstrap p-value is the Monte Carlo-corrected two-sided sign-tail probability, and the interval is percentile-based. HAC and bootstrap are not independent replications of evidence: they are two uncertainty approximations for one coefficient. The conjunctive label is therefore a pre-specified robustness convention, not a theorem that the joint false-discovery rate is exactly five percent. Benjamini–Hochberg's formal guarantee also requires dependence conditions that need not hold for the four correlated summaries; raw and adjusted values and the full proxy correlation matrix will be reported so this limitation is visible.

Event-window interactions for Volmageddon, COVID, and the August 2024 yen-carry episode are secondary heterogeneity analyses. They are not used to select the sample or to validate a desired sign. Absolute return is a secondary outcome. The IV-surface/RL tournament and critical-slowing-down exercises from the initial registration are exploratory diagnostics, not confirmatory evidence for the mechanism.

7.2 Ex-ante precision calculation

The protocol does not promise that the available sample can detect every economically interesting association. To expose the relevant scale before extraction, residualize a standardized proxy Q on the controls and let R^2_Q be the auxiliary regression's R^2. Under an idealized Gaussian approximation with effective independent sample size n_{eff}, residual outcome standard deviation σ_e, two-sided five-percent size, and 80-percent power, the minimum detectable coefficient is approximately

$$\frac{|b|_{\text{MDE}}}{\sigma_e} \simeq \frac{z_{0.975} + z_{0.80}}{\sqrt{n_{\text{eff}}(1 - R^2_Q)}} = \frac{2.802}{\sqrt{n_{\text{eff}}(1 - R^2_Q)}}. \tag{23}$$

The actual n_{eff}, R^2_Q, and σ_e are unknown before extraction. They will be reported, not selected. Because no smallest effect of interest was specified, a null result supports only "not detected under this design"; it cannot establish practical equivalence to zero.

n_{eff}	$R_Q^2 = 0$	$R_Q^2 = 0.5$
500	$0.125\,\sigma_e$	$0.177\,\sigma_e$
1000	$0.089\,\sigma_e$	$0.125\,\sigma_e$
1500	$0.072\,\sigma_e$	$0.102\,\sigma_e$
2000	$0.063\,\sigma_e$	$0.089\,\sigma_e$

Table 4: Idealized 80-percent-power sensitivity from eq. (23). Serial dependence, heavy tails, missing rows, collinearity, and multiplicity reduce effective precision; the table is not a claim about realized power.

7.3 Decision language fixed before extraction

The following interpretation is part of the design.

- If no registered book summary has an incremental association, predictive relevance at the daily horizon is not detected under these measurements. Without an equivalence margin, failure to reject is not evidence that every economically meaningful effect is absent.
- An association for U_t, B_t, M_t, or D_t supports predictive content in an observable option-positioning summary, but does not identify dealer sign or the structural feedback in (1).
- Results that depend on a sign convention are labelled non-identified. They cannot be reported as confirmation that dealers were long or short gamma.
- Even a robust predictive association is not causal. A direct test of the signed-book mechanism requires participant-tagged positions or another independently validated dealer- inventory measure, plus an identification strategy for endogenous option demand.

This hierarchy prevents a proxy regression from being upgraded into a structural test after its sign is known.

8 Limitations and falsifiable scope

The analysis has substantial limitations.

1. **Local deterministic result.** The Hopf theorem concerns the drift skeleton near one fixed equilibrium. Brownian sample paths are not periodic orbits, and no global attractor, ergodic distribution, or tail statement is proved.
2. **Reduced-form price pressure.** The sign and scale of κ are not derived from dealer optimization, market clearing, or Gârleanu–Pedersen–Poteshman. A microfounded equilibrium model could reject the chosen state equations.
3. **Frozen low-dimensional book.** Actual positions evolve by strike, maturity, and participant. A single Gaussian density and one effective maturity suppress these dynamics. The analytic kernel also identifies the physical variance state with a flat Black–Scholes variance, omitting the implied-volatility surface and variance-risk-premium wedge. The mixture audit shows that nonlinear classification can change even when the linear Hopf root survives. The phase map is consequently illustrative.

4. **Variance boundary and explosion.** The added variance drift points inward at zero, but this paper does not prove global non-explosion for every parameter and initial condition. The theorem and numerical example are local.

5. **Measurement.** Public OI does not recover signed dealer inventory. September WRDS access will improve coverage and reproducibility, not solve this identification problem.

6. **Theory–proxy gap.** The theorem is written for a latent signed book; the primary future regressors are unsigned summaries. A predictive association for those summaries is not a direct test of the theorem's sign channel, and a null association does not refute every possible participant-level signed-book mechanism.

7. **Predictive specification.** Log squared return is noisy and depends on the fixed 10^{-8} floor at very small returns. The controls may be collinear, HAC and block bootstrap rely on weak-dependence approximations, and four correlated proxy tests complicate literal FDR interpretation. Absolute return is reported as a pre-specified secondary outcome, not substituted if it is more favourable.

8. **Retrospective registered horizon.** The 2017–2024 period and its stress episodes were historically observable when the analysis plan was written. The timestamp prevents adaptation to the unextracted proprietary option proxies; it does not make the historical outcome path unknown or turn the design into a prospective experiment. The end date is inherited from the registered horizon and is not the last vendor date available in 2026.

9. **No causal identification.** Option demand, inventory, volatility expectations, and underlying returns are jointly determined. Forward dating removes mechanical look-ahead but not omitted variables, anticipation, measurement error, or equilibrium endogeneity.

10. **End-of-day horizon.** Contracts expiring on date t are excluded because they have no life after the close, and vendor OI can reflect the preceding clearing cycle. The registered next-day design therefore cannot test intraday 0DTE feedback, even though that segment may be economically important.

11. **No stochastic-threshold result.** For the frozen affine additive surrogate, the tangent exponent is the spectral abscissa of J, independent of additive noise scale. The full square-root model has state-dependent diffusion and a multiplicative stochastic variational equation. Nonlinear random dynamical systems can have genuinely stochastic bifurcations (Baxendale, 1994, 2025), but that requires a different analysis.

12. **No Hawkes equivalence.** Near-critical Hawkes limits (Jaisson and Rosenbaum, 2015) concern a branching structure. They neither identify the states in (1)–(3) nor imply a complex eigenvalue pair.

These restrictions make the paper's claim smaller but testable: a specified local feedback system can undergo a non-degenerate Hopf bifurcation under explicit conditions, and one transparent nonlinear example satisfies them. Whether option-market dealer positions generate a related mechanism in data remains open.

9 Reproducibility and conclusion

The source, analytic derivatives, numerical experiment, and tests are available at `https://github.com/mahimn01/reflexive-options`. The principal checks independently verify that the equilibrium does not move with κ, the variance-boundary drift is inward for arbitrary memory values, the quadratic (14) matches the direct characteristic polynomial, the stated eigenvalues occur at the reported root, and the nonlinear orbit keeps $v > 0$. The arXiv bundle contains only referenced assets and is built from a clean extraction test.

The main theoretical lesson is simple. A three-state option-feedback model can support a local volatility cycle, but only after the equilibrium family, state-space boundary, nonlinear derivatives, and Routh–Hurwitz side conditions are handled consistently. The main empirical lesson is equally important: open interest is not dealer inventory. The future data exercise is therefore designed to report what the proxy predicts and to stop where identification stops.

A Derivation of the Gaussian book integral

Normalize the reference forward $F_\star$ to one, write $S = e^X$, and let $k = \log K$. Black–Scholes spot gamma for a common maturity T is

$$\Gamma^{BS}(X, k, v, T) = \frac{e^{-q_d T - X}}{\sqrt{vT}} \phi\left(\frac{X - k + (r - q_d + v/2)T}{\sqrt{vT}}\right), \tag{24}$$

where ϕ is the standard-normal density. With $k \sim N(\mu_q, \sigma_q^2)$, the positive unit-mass book is

$$\begin{aligned} \mathcal{G}(X, v) &= \int_{\mathbb{R}} \varphi_q(k) \Gamma^{BS}(X, k, v, T)\,\mathrm{d}k \\ &= e^{-q_d T - X} \int_{\mathbb{R}} \varphi(k; \mu_q, \sigma_q^2) \varphi\{k; X + (r - q_d + v/2)T, vT\}\,\mathrm{d}k. \end{aligned} \tag{25}$$

The integral of two Gaussian densities in the same variable is the Gaussian density of the difference of their means with variance equal to the sum:

$$\int \varphi(k; \mu_1, s_1^2) \varphi(k; \mu_2, s_2^2)\,\mathrm{d}k = \varphi(\mu_1; \mu_2, s_1^2 + s_2^2).$$

Applying the identity to eq. (25), setting $\tau^2(v) = \sigma_q^2 + vT$ and $m(v) = \mu_q - (r - q_d + v/2)T$, gives eq. (4). A direct completion of the square yields the same expression. Because $\sigma_q > 0$, the convolved formula is smooth in v on a neighbourhood of every $\theta_v > 0$.

This derivation uses a fixed strike distribution. The dependence on X arises because a price movement changes the moneyness of those fixed strikes; it is not an assumption that positions instantly move with spot. Gross contract mass, contract multiplier, and a price- impact coefficient are omitted because proposition 2 shows that a common positive scale is observationally inseparable from κ in this local model.

B Analytic derivative audit

Let $D = X - m(v)$, $t = \tau^2(v)$, and $h = \log \mathcal{G}$. Constants independent of (X, v) can be suppressed, leaving

$$h = -X - \tfrac{1}{2} \log t - \frac{D^2}{2t}, \qquad D_v = T/2, \qquad t_v = T.$$

The log-derivatives needed through order three are

$$
\begin{aligned}
h_X &= -1 - D/t, & h_{XX} &= -1/t, & h_{XXX} &= 0,\\
h_v &= -\frac{T}{2t} - \frac{DT}{2t} + \frac{D^2T}{2t^2}, \\
h_{vv} &= \frac{T^2}{2t^2} - \frac{T^2}{4t} + \frac{DT^2}{t^2} - \frac{D^2T^2}{t^3}, \\
h_{vvv} &= -\frac{T^3}{t^3} + \frac{3T^3}{4t^2} - \frac{3DT^3}{t^3} + \frac{3D^2T^3}{t^4}, \\
h_{Xv} &= -\frac{T}{2t} + \frac{DT}{t^2}, & h_{XXv} &= \frac{T}{t^2}, & h_{Xvv} &= \frac{T^2}{t^2} - \frac{2DT^2}{t^3}.
\end{aligned}
\tag{26}
$$

For indices $i, j, k \in \{X, v\}$, raw derivatives follow from

$$
\begin{aligned}
\mathcal{G}_i &= \mathcal{G}h_i, \\
\mathcal{G}_{ij} &= \mathcal{G}(h_ih_j + h_{ij}), \\
\mathcal{G}_{ijk} &= \mathcal{G}(h_ih_jh_k + h_ih_{jk} + h_jh_{ik} + h_kh_{ij} + h_{ijk}).
\end{aligned}
\tag{27}
$$

The centered constant in eq. (5) has zero derivative, so every non-zero derivative of g is $s\mathcal{G}_{i\cdots}/\mathcal{G}(0, \theta_v)$.

At the canonical configuration, the complete non-zero book derivative set used in B and C is

Derivative	Value	Derivative	Value	Derivative	Value
G_X	−0.0617978	G_v	−2.0198878	G_{XX}	−17.9737091
G_{Xv}	−6.3390058	G_{vv}	11.8556078	G_{XXX}	3.3326765
G_{XXv}	117.9016552	G_{Xvv}	83.4816197	G_{vvv}	−114.8072302

Table 5: Analytic normalized derivatives at $(0, \theta_v)$. Symbolic differentiation of eq. (4) agrees to displayed precision.

For the drift expansion $f(z_\star + x) = Jx + \frac{1}{2}B(x, x) + \frac{1}{6}C(x, x, x) + O(\|x\|^4)$, the book contributes

$$
\begin{aligned}
&B_{1,XX} = \kappa G_{XX}, && B_{1,Xv} = B_{1,vX} = \kappa G_{Xv}, && B_{1,vv} = \kappa G_{vv}, \\
&C_{1,XXX} = \kappa G_{XXX}, && C_{1,XXv} = \kappa G_{XXv}, && C_{1,Xvv} = \kappa G_{Xvv}, \quad C_{1,vvv} = \kappa G_{vvv},
\end{aligned}
$$

with all permutations of the last three indices filled symmetrically. The variance drift adds $B_{2,v\chi} = B_{2,\chi v} = \gamma$. There are no other quadratic or cubic drift derivatives in the static Gaussian specialization.

C Algebraic and numerical audit trail

For completeness, with $G_\chi = 0$ the coefficients are

$$
\begin{aligned}
c_2 &= A + \delta - \kappa G_X, \\
c_1 &= M + \delta A - \kappa G_X A, \\
c_0 &= \delta M + \tfrac{1}{2}L - \kappa(G_X M + G_v L).
\end{aligned}
$$

Multiplying the first two and subtracting the third gives

$$
\begin{aligned}
H(\kappa) = G_X^2 A\kappa^2 &+ \{G_v L - G_X A(A + 2\delta)\}\kappa \\
&+ A(M + \delta A + \delta^2) - \tfrac{1}{2}L,
\end{aligned}
$$

which is (15). If $G_X = 0$, the equation is linear unless $G_v L = 0$; the implementation handles this degeneracy separately. If the discriminant is non-negative, both real roots are enumerated in ascending order and each is subjected to the positivity, frequency, factorization, and transversality checks. Thus the reported root is not selected by sign alone.

At a valid root, factorization gives $P_\lambda(i\omega) = 2\{-c_1 + ic_2\sqrt{c_1}\}$. Combining this with $P_\kappa(i\omega)$, using $c_0 = c_1 c_2$, and taking the real part of $-P_\kappa/P_\lambda$ yields (13). This also shows why $H' \neq 0$ is the relevant crossing condition but does not, by itself, determine which side contains stable cycles; that requires the sign of both (13) and ℓ_1.

C.1 Root-classification algorithm

The implementation follows the following deterministic sequence.

1. Compute analytic G_X, G_v and (A_2, A_1, A_0).
2. If $|A_2|$ is below tolerance, solve the non-degenerate linear equation; otherwise compute the quadratic discriminant and enumerate both real roots.
3. Sort positive roots. For each root, rebuild J directly and compute the characteristic polynomial numerically rather than reusing the quadratic coefficients.
4. Require $c_2, c_1, c_0 > 0$, a scaled residual $|c_1 c_2 - c_0| \leq 10^{-7} \max(1, |c_0|, |c_1 c_2|)$, non-zero frequency, eigenvalue factorization, and non-zero transversality.
5. Construct analytic B, C and evaluate eq. (17) at every root passing the linear checks. Return all valid points in ascending order. Analyses explicitly labelled "threshold" use the first point; otherwise all points are reported. If none passes, report that no valid positive local Hopf root was found.

The algorithm never converts a negative discriminant into a stability conclusion.

C.2 Nonlinear convergence table

$100\varepsilon_\kappa$	Transient horizon	A_X	A_v	Period	v range
0.0625	2269.50	0.00442	0.00325	0.13341	$[0.05927, 0.06577]$
0.125	1134.75	0.00626	0.00460	0.13347	$[0.05795, 0.06715]$
0.25	567.37	0.00887	0.00651	0.13361	$[0.05608, 0.06910]$
0.5	283.69	0.01259	0.00922	0.13389	$[0.05346, 0.07191]$
1.0	141.84	0.01795	0.01309	0.13450	$[0.04978, 0.07595]$
2.0	80.00	0.02586	0.01867	0.13596	$[0.04453, 0.08187]$
4.0	80.00	0.03844	0.02713	0.14061	$[0.03664, 0.09089]$
6.0	80.00	0.05337	0.03625	0.15667	$[0.02769, 0.10018]$

Table 6: Actual-book deterministic cycles. A_X and A_v are half peak-to-peak amplitudes measured over the final five simulated years. All times are in years.

The solver is DOP853 with default audit tolerances $(10^{-9}, 10^{-11})$ and maximum step 0.005 years. Periods are averages of successive local maxima of X over the retained window. The plotted $1.02\kappa^\star$ orbit in fig. 1 is the same configuration as the middle row of table 6; the earlier 40-year discard is more than sufficient for the plotted initialization, while the robustness audit uses the more conservative 80-year rule.

D Empirical implementation and reporting audit

The data workflow is ordered to prevent outcome knowledge from influencing contract construction.

1. Archive vendor documentation, query text, raw extracts, retrieval time, row count, date range, and SHA-256 hashes. Do not construct returns at this stage.
2. Verify SPX security identifiers, OSI roots, adjustment and settlement fields, contract multipliers, unique option identifiers, and the OptionMetrics spot mapping. Non-identical duplicates are a no-go error rather than an invitation to choose a row.
3. Apply every contract filter in A14 and record sequential attrition: input rows, root/security, adjustment/multiplier, settlement, DTE, quote, IV, duplicate, forward, and moneyness stages. Record zero-OI rows even if later dropped for speed.
4. Construct parity forwards within date–expiration–settlement groups. Record the fraction of gamma mass using carry fallback and zero-curve endpoint extension. Compare recomputed gamma with vendor gamma only as a mapping diagnostic.
5. Construct (U, B, M, D) and freeze the common finite option-date sample. No daily summary is winsorized, clipped, or removed because it is extreme.
6. Only after the option-side freeze, archive CRSP returns and official Cboe VIX. Build the complete ordered CRSP trading-session calendar, construct return leads and lags before removing any date with a missing option proxy, use outcome-session weekday indicators, and record every listwise deletion as fixed in A15.
7. Run all four primary equations with the same controls and aligned rows. Save the design-matrix hashes, coefficients, HAC covariance matrices, bootstrap seeds and block indices, percentile distributions, VIFs, proxy correlations, and adjusted p-values.
8. Produce the full primary table before any exploratory event, convention-signed, CSD, or RL analysis is opened.

For transparency, the Benjamini–Hochberg adjusted value for ordered raw values $p_{(1)} \leq \cdots \leq p_{(m)}$ is

$$\widetilde{p}_{(i)} = \min_{j \geq i} \min\{1, mp_{(j)}/j\},$$

mapped back to the original proxy order. HAC and bootstrap families are adjusted separately as locked in A14. The moving-block bootstrap draws starting indices uniformly from all overlapping length-10 blocks, concatenates enough blocks to obtain n rows, and truncates the last block. The same starts are used for every primary proxy.

The final empirical report must include, even when inconvenient: the complete attrition table; excluded dates and reasons; parity-fallback mass; descriptive distributions in raw units; the four-by-four proxy correlation matrix; VIFs; coefficient, HAC standard error and raw/adjusted p-value; bootstrap standard error, raw/adjusted p-value and interval; pre-specified label; all signed-convention sensitivities; and a deviations table comparing every post-data change with the registered analysis. Participant-level or causal language is prohibited unless new data and a new identification design genuinely support it.

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